Hyperbolic Diffusion

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Suppose that we run hyperbolic Brownian motion on the half-space model in n dimensions. Suppose we are concerned with diffusing starting at the origin $\mathbf{0}$ in infinite time to one of a finite vocabulary $V \subset \partial\mathbb{H}$. From any finite point $z \in \mathbb{H}$, the probability density in the diffusion process conditioned on approahing some point in V of going to a particular point $x \in V$ is

$$q(x \mid z) = \frac{\exp\left((n-1)\langle x, z\rangle_{\mathbb{H}}\right)}{\sum_{v \in V} \exp\left((n-1)\langle v, z\rangle_{\mathbb{H}}\right)},$$

where $\langle v, z\rangle_{\mathbb{H}}$ denotes the signed distance to a horosphere through $\mathbf{0}$ with limiting boundary point $v \in \partial\mathbb{H}$. In the ball model, iirc,

$$\langle b, z\rangle_{\mathbb{H}} = \log\left(1 - \|z\|^2\right) - \log\left(\|b - z\|^2\right).$$

We can derive this as follows:

$$\begin{aligned}
\langle b, z\rangle_{\mathbb{H}} &= \lim_{\gamma \to 1_-} d_{\mathbb{H}}(\gamma b, 0) - d_{\mathbb{H}}(\gamma b, z) \\
&= \lim_{\gamma \to 1_-} 2\log\left(\frac{\|0 - \gamma b\| + \sqrt{\|0\|^2 \|\gamma b\|^2 - 2\gamma b^T 0 + 1}}{\sqrt{(1 - \|0\|^2)(1 - \|\gamma b\|^2)}}\right) - 2\log\left(\frac{\|z - \gamma b\| + \sqrt{\|z\|^2 \|\gamma b\|^2 - 2\gamma b^T z + 1}}{\sqrt{(1 - \|z\|^2)(1 - \|\gamma b\|^2)}}\right) \\
&= \lim_{\gamma \to 1_-} 2\log\left(\frac{\gamma + 1}{\sqrt{1 - \gamma^2}}\right) - 2\log\left(\frac{\|z - \gamma b\| + \sqrt{\gamma^2 \|z\|^2 - 2\gamma b^T z + \|b\|^2}}{\sqrt{(1 - \|z\|^2)(1 - \gamma^2)}}\right) \\
&= \lim_{\gamma \to 1_-} -2\log\left(\frac{\|z - \gamma b\| + \|\gamma z - b\|}{(\gamma + 1)\sqrt{1 - \|z\|^2}}\right) \\
&= -2\log\left(\frac{2\|b - z\|}{2\sqrt{1 - \|z\|^2}}\right) \\
&= \log\left(1 - \|z\|^2\right) - \log\left(\|b - z\|^2\right).
\end{aligned}$$

Now, think about making a small step of Brownian motion in hyperbolic space with timestep δ. Locally, this is just

$$P(z_{t+\delta} \mid z_t) \propto \exp\left(-\frac{\|z_{t+\delta} - z_t\|^2}{2\delta}\right).$$

Now what if we condition on the event that $z_\infty \in V$. Of course,

$$\begin{aligned}
P(z_{t+\delta} \mid z_t, z_{\infty \in V}) &= \frac{P(z_{\infty\in V} \mid z_{t+\delta}) \cdot P(z_{t+\delta} \mid z_t)}{P(z_{\infty \in V} \mid z_t)} \\
&\propto \frac{\sum_{v\in V} \exp\left((n-1)\langle v, z_{t+\delta}\rangle_{\mathbb{H}}\right)}{\sum_{v\in V} \exp\left((n-1)\langle v, z_t\rangle_{\mathbb{H}}\right)} \exp\left(-\frac{\|z_{t+\delta} - z_t\|^2}{2\delta}\right) \\
&\propto \exp\left(-\frac{\|z_{t+\delta} - z_t\|^2}{2\delta}\right) \sum_{v\in V} \exp\left((n-1)\langle v, z_{t+\delta}\rangle_{\mathbb{H}}\right) \\
&\approx \exp\left(-\frac{\|z_{t+\delta} - z_t\|^2}{2\delta}\right) \sum_{v\in V} \exp\left((n-1)\left(\langle v, z_t\rangle_{\mathbb{H}} + \langle z_{t+\delta} - z_t, (z \to v)\rangle\right)\right) \\
&= \sum_{v\in V} \exp\left((n-1)\langle v, z_t\rangle_{\mathbb{H}}\right) \cdot \exp\left(-\frac{\|z_{t+\delta} - z_t\|^2}{2\delta} + (n-1)\langle z_{t+\delta} - z_t, (z \to v)\rangle\right) \\
&\propto \sum_{v\in V} \exp\left((n-1)\langle v, z_t\rangle_{\mathbb{H}}\right) \cdot \exp\left(-\frac{\|z_{t+\delta} - z_t - (n-1)\cdot\delta\cdot(z\to v)\|^2}{2\delta}\right).
\end{aligned}$$

This is of course a mixture model with mean $z_t + (n-1)\cdot\delta\cdot(z \to v)$ chosen with probability proportional to $\exp\left((n-1)\langle v, z_t\rangle_{\mathbb{H}}\right)$. For small δ this mixture model becomes just a Gaussian with mean equal to the mean of the mixture. This means that the whole thing is just

$$P(z_{t+\delta} \mid z_t, z_{\infty\in V}) = \mathcal{N}\left((n-1)\cdot\delta\cdot\frac{\sum_{v\in V}\exp\left((n-1)\langle v, z_t\rangle_{\mathbb{H}}\right)\cdot(z\to v)}{\sum_{v\in V}\exp\left((n-1)\langle v, z_t\rangle_{\mathbb{H}}\right)}, \delta I\right)$$

For the forward noising process, we diffuse from input $x \in V$ (at time $t = \infty$) to $\mathbf{0}$ (at time $t = 0$) via hyperbolic Brownian motion conditioned to those endpoints. Obviously,

$$\begin{aligned}
r_{t|0,\infty}(z \mid \mathbf{0},\ x) &= \frac{r_{t|0}(z \mid \mathbf{0}) r_{\infty|t}(x \mid z)}{r_{\infty|0}(x \mid \mathbf{0})} \\
&= \rho_t(d_{\mathbb{H}}(z, \mathbf{0})) \cdot \exp\left((n-1)\langle x, z\rangle_{\mathbb{H}}\right).
\end{aligned}$$

This is the forward process. Let the reverse denoising process have

$$p(x \mid z) = \frac{\exp\left(f_\theta(x; z) + (n-1)\langle x, z\rangle_{\mathbb{H}}\right)}{\sum_{v\in V}\exp\left(f_\theta(v; z) + (n-1)\langle v, z\rangle_{\mathbb{H}}\right)}.$$

Equivalently, the reverse SDE (in hyperbolic space) can be

$$\dot{z} = \dot{W} + (n-1)\frac{\sum_{x\in V}\exp\left(f_\theta(x; z) + (n-1)\langle x, z\rangle_{\mathbb{H}}\right)\cdot(z\to x)}{\sum_{v\in V}\exp\left(f_\theta(v; z) + (n-1)\langle v, z\rangle_{\mathbb{H}}\right)}$$

where $(z \to x)$ denotes the unit vector in the tangent space of $\mathbb{H}$ at z pointing in the direction of x. Note that $(z \to x)$ is the gradient with respect to z of $\langle x, z\rangle_{\mathbb{H}}$ at z. In particular, recall that

$$\begin{aligned}
\nabla_z \langle b, z\rangle_{\mathbb{H}} &= \nabla_z\left(\log\left(1 - \|z\|^2\right) - \log\left(\|b - z\|^2\right)\right) \\
&= -\frac{2z}{1 - \|z\|^2} - \frac{2(z-b)}{\|z-b\|^2}.
\end{aligned}$$

Observe that

$$\left\| -\frac{2z}{1-\|z\|^2} - \frac{2(z-b)}{\|z-b\|^2} \right\|^2 = \frac{4}{(1-\|z\|^2)^2}.$$

As expected, this is exactly the scale factor of the metric tensor of the Poincare ball model! In the tangent space, then, the unit vector $(z \to b)$ is

$$
\begin{aligned}
(z \to b) &= \left(-\frac{2z}{1 - \|z\|^2} - \frac{2(z-b)}{\|z-b\|^2} \right) \cdot \left(\frac{4}{(1-\|z\|^2)^2} \right)^{-1/2} \\
&= \left(-\frac{2z}{1 - \|z\|^2} - \frac{2(z-b)}{\|z-b\|^2} \right) \cdot \left(\frac{1-\|z\|^2}{2} \right) \\
&= (b-z)\frac{1-\|z\|^2}{\|z-b\|^2} - z \\
&= (b-z)\exp(\langle b, z\rangle_{\mathbb{H}}) - z.
\end{aligned}
$$

The actual loss is then proportional to

$$
\left\| \frac{\sum_{v\in V} \exp\left(f_\theta(v;z) + (n-1)\langle v, z\rangle_{\mathbb{H}}\right) \cdot (z \to v)}{\sum_{u\in V} \exp\left(f_\theta(u;z) + (n-1)\langle u, z\rangle_{\mathbb{H}}\right)} - (z \to x) \right\|^2 = \left\| \mathbf{E}_{v\sim\mu_\theta(z)}\left[(z \to v) - (z \to x)\right] \right\|^2
$$

As a function of time, consider the target size

$$
\left\| \frac{\sum_{v\in V} \exp\left((n-1)\langle v, z\rangle_{\mathbb{H}}\right) \cdot (z \to v)}{\sum_{u\in V} \exp\left((n-1)\langle u, z\rangle_{\mathbb{H}}\right)} - (z \to x) \right\|^2.
$$

This is a reasonable proxy we can use for doing importance sampling in time.

More intuitively, we can describe the processes as follows. The *noising process* (for a single token) is just an infinite-time hyperbolic Brownian bridge from the boundary point x (the example) to the origin 0. The *discretized denoising process* does the following at each timestep:

- Let $z \in \mathbb{H}$ be the point we are currently at.
- For each possible embedding point v in the vocabulary (on the boundary), compute the logits of the "prior" distribution of going to v conditioned on being at x. Those logits are

$$
(n-1)\langle v, z\rangle_{\mathbb{H}}.
$$

- Add to those logits the prediction logits f_θ from the learned model.
- Sample from the resulting distribution (i.e. the multinomial distribution with those logits), producing some point b (on the boundary).
- Simulate an infinite-time hyperbolic Brownian bridge process starting at position z and approaching b for Δ time.
- The new state is the result z' of this simulation.

The actual *denoising process* is the continuous limit of this process as the timestep Δ approaches 0.

1 Learning

The NELBO for this is going to be

$$
\mathrm{NELBO}_\theta(x) = (n-1)^2 \int_0^\infty \mathbf{E}_{z\sim q_t(\cdot|x)} \left[\left\| \sum_{v\in V} \frac{\exp\left(f_\theta(v;z) + (n-1)\langle v, z\rangle_{\mathbb{H}}\right)}{\sum_{u\in V} \exp\left(f_\theta(u;z) + (n-1)\langle u, z\rangle_{\mathbb{H}}\right)} \cdot \left((z \to v) - (z \to x)\right) \right\|^2 \right] dt
$$

We should be able to parameterize this as an expected value over (1) a random distance from the origin, (2) a random angle between z and x as a function of that random distance, and (3) a random rotation to handle the rest of the dimensions. One dumb way to handle this is to let z be the result of taking a random vector, scaling it so it has the right magnitude, and then reflecting it so that it has the correct angle. But that isn't continuous... There's gotta be some way we can write this so we can backprop through the embeddings and thereby actually jointly learn the embeddings!

Here's one way. First, draw a two-dimensional value at random based on t, ranging over the right half-plane. Then apply a uniform random rotation. Finally, reflect so that e_1 maps to z.

In particular, when $t = 0$, $z = 0$ and

$$\begin{aligned}
\mathrm{NELBO}_\theta(x;0) &= (n-1)^2 \left\| \sum_{v \in V} \frac{\exp\left(f_\theta(v;0) + (n-1)\langle v, 0\rangle_{\mathbb{H}}\right)}{\sum_{u \in V} \exp\left(f_\theta(u;0) + (n-1)\langle u, 0\rangle_{\mathbb{H}}\right)} \cdot ((0 \to v) - (0 \to x)) \right\|^2 \\
&= (n-1)^2 \left\| \sum_{v \in V} \frac{\exp\left(f_\theta(v;0)\right)}{\sum_{u \in V} \exp\left(f_\theta(u;0)\right)} \cdot (v - x) \right\|^2 \\
&= (n-1)^2 \left\| \sum_{v \in V} \mu_v \cdot (v - x) \right\|^2 .
\end{aligned}$$

This is minimized over the distribution when the prior distribution is equal to the marginal distribution of each token of x.

$$\frac{\partial}{\partial \mu_u} \mathrm{NELBO}_\theta(x;0) = 2(n-1)^2 \cdot (u - x)^T \left(\sum_{v \in V} \mu_v \cdot (v - x) \right)$$